

INTERNSHIP LEARNING REPORT

Introduction

During my internship at the tractor manufacturing plant, I learned about various departments, industrial equipment, quality inspection methods, maintenance practices, and manufacturing processes. I observed how different departments work together to manufacture tractors efficiently while maintaining quality and safety. I also gained practical knowledge about diesel generators, metrology laboratory instruments, gauges, air compressors, engine assembly, and preventive maintenance.

Plant Departments

During the internship, I visited the following departments:

1. Security Department

Function: Controls the entry and exit of employees, visitors, and materials. Ensures the safety and security of the factory.

2. Stores/Warehouse

Function: Receives, stores, and issues raw materials and spare parts required for production.

3. Vendor Quality (Receiving Inspection)

Function: Inspects incoming materials from suppliers to ensure they meet quality standards before use.

4. Metrology Laboratory

Function: Measures and inspects components using precision instruments to ensure dimensional accuracy.

5. Engine Assembly Department

Function: Assembles engine parts such as the crankshaft, piston, cylinder head, and other components to build a complete engine.

6. Engine Testing Department

Function: Tests the assembled engine for performance, leakage, noise, and proper operation before it is sent to the assembly line.

7. Tractor Assembly Department

Function: Assembles the engine, transmission, hydraulic system, electrical system, and tyres to manufacture a complete tractor.

8. Maintenance Department

Function: Maintains machines and equipment to reduce breakdowns and ensure smooth production.

9. Purchase Department

Function: Purchases raw materials, machine parts, and other items required for manufacturing.

10. Manufacturing Department

Function: Produces tractor components from raw materials through machining, welding, and fabrication processes.

11. Production Office

Function: Plans and monitors daily production activities to achieve production targets.

12. Design (R&D) Department

Function: Designs new tractor models and improves existing products.

13. Finance Department

Function: Manages the company's budget, expenses, and financial planning.

14. Accounts Department

Function: Maintains financial records, prepares invoices, and records all business transactions.

15. Sales and Marketing Department

Function: Promotes tractor sales, manages dealers, and maintains customer relationships.

16. Logistics Department

Function: Manages the transportation, storage, and delivery of materials and finished tractors.

This order follows the typical workflow you observed during your internship and is suitable for inclusion in your report.

1.Tractor plant process

Maintenance and Utility Sub Station

- **500 kVA Diesel Generator (DG Set)**

The 500 kVA Diesel Generator is used as a backup power source during power failures.

Specifications

- Capacity: 500 kVA
- Output Power: 400 kW
- Supply: Three-phase AC
- Frequency: 50 Hz

Working

- The DG set starts automatically or manually during power failure.
- The diesel engine rotates the alternator.
- The alternator converts mechanical energy into electrical energy.
- The generated electricity is supplied to the industrial electrical distribution panel.

Maintenance

- Check engine oil level.
- Inspect fuel line leakage.
- Check battery condition.
- Clean air filters.
- Inspect coolant level regularly.

- **900-Litre Diesel Day Tank**

The diesel day tank stores fuel required for daily generator operation.

Purpose

- Stores diesel for daily operation.
- Capacity: 900 litres.
- Ensures continuous fuel supply to the DG set.

Working

- Fuel is supplied to the generator through fuel pipelines.
- Excess fuel returns to the day tank through the return line

Air Compressor

An air compressor converts atmospheric air into high-pressure compressed air.

Main Components

- Air Filter
- Screw Compressor Element
- Air Receiver Tank
- Oil Separator
- Oil Cooler
- Air Cooler
- Oil Filter
- Safety Valve

Applications

Compressed air is used for pneumatic tools, machine operation, cleaning, painting, and automation in the manufacturing plant.

● Assembly Section

1. Chassis Assembly

Function: The tractor frame (chassis) is prepared as the base structure for mounting all major components.

2. Engine Assembly

Function: The assembled and tested engine is mounted onto the chassis and securely fastened.

3. Transmission Assembly

Function: The clutch, gearbox, and transmission system are installed to transmit power from the engine to the wheels.

4. Front Axle Assembly

Function: The front axle and steering components are fitted to provide steering and support the front wheels.

5. Rear Axle Assembly

Function: The rear axle and differential are installed to transmit power to the rear wheels.

6. Hydraulic Assembly

Function: Hydraulic pumps, lift arms, cylinders, and pipes are installed for lifting and operating farm implements.

7. Electrical Assembly

Function: The wiring harness, battery, starter motor, alternator, lights, and switches are connected and tested.

8. Fuel System Assembly

Function: The fuel tank, fuel lines, filters, and injectors are installed to supply fuel to the engine.

9. Tyre and Wheel Assembly

Function: Front and rear tyres are mounted, and wheel nuts are tightened to the specified torque.

10. Final Inspection and Testing

Function: The complete tractor is inspected for quality, tested for engine performance, brakes, steering, hydraulics, and electrical systems before dispatch.

Engine Plant Process

The engine plant consists of the following sections:

- Security
- Stores
- Vendor Quality (Receiving Inspection)
- Metrology Laboratory
- Engine Assembly
- Engine Testing
- Maintenance

- Diesel Storage

Metrology Laboratory

The Metrology Laboratory is responsible for measuring and inspecting components accurately.

Instruments Learned

- Coordinate Measuring Machine (CMM)
- Spectrometer (Material Composition Test)
- Particle Count Analyzer
- Metallurgical Microscope
- Vickers Hardness Tester
- Rockwell Hardness Tester
- Brinell Hardness Tester
- Universal Testing Machine (UTM)
- Coating Thickness Meter
- Surface Roughness Tester
- Crack Detector
- Salt Spray Test Equipment
- Magnetic Particle Inspection (MPI)
- Specific Gravity Testing Equipment

Purpose

These instruments are used to measure dimensions, hardness, surface finish, coating thickness, material composition, and detect cracks to ensure product quality.

Gauges Used in Industry

Types of Gauges

During the internship, I learned about different gauges used for inspection.

- Plug Gauge – Checks internal diameter.
- Ring Gauge – Checks external diameter (Go/No-Go).
- Snap Gauge – Measures outside diameter quickly.
- Thread Plug Gauge – Checks internal threads.
- Thread Ring Gauge – Checks external threads.
- Vernier Caliper – Measures length, width, and depth.
- Micrometer – Measures thickness with high accuracy.

- Height Gauge – Measures vertical dimensions.
- Feeler Gauge – Measures gap or clearance.
- Dial Gauge – Measures run-out and flatness.
- Bore Dial Gauge – Measures bore diameter.
- Profile Projector – Measures angles, radius, and profile using magnification.
- Depth Gauge – Measures hole depth.